Step 1: Define the workload

The first step in the WAR process is to define the workload. This includes identifying the applications, databases, and other components that make up the workload, as well as the dependencies between these components.

Step 2: Define the core team

The core team should include individuals who have a deep understanding of the workload and its architecture. This may include architects, developers, operations personnel, and others.

Step 3: Decide on the AWS pillars and lenses

Based on the nature of the workload, the team will decide which of the AWS pillars and lenses are most relevant. For example, if the workload includes a web application, the team may focus on the security and performance efficiency pillars.

Step 4: Conduct the review

The team will then use the AWS well-architected tool to conduct the review. This involves answering a series of questions related to the chosen pillars and lenses, and identifying any high-risk issues (HRIs).

Step 5: Create an improvement plan

Based on the results of the review, the team will create an improvement plan to address the identified HRIs. This may involve making changes to the architecture, implementing new AWS services or features, or modifying operational procedures.

Best practices for implementing AWS WAR

Well-Architected Reviews should be leveraged for rock-solid AWS deployment. Let's see how this can be done.

- ***Harness the free tool:*** The AWS well-architected tool benchmarks the setup against best practices. Think of it as a free security and efficiency check-up for your cloud infrastructure.
- ***Six pillars, one goal:*** WAR systematically assesses these pillars to identify areas for improvement. Imagine reviewing your cloud environment from six different angles to ensure optimal health.
- ***Focus on the big threats:*** The review prioritises potential risks. For example, it may reveal an overly permissive IAM policy (security pillar). These high-risk issues can be addressed first.
- ***Collaboration is key:*** The review isn't a one-sided audit. Developers and business stakeholders are involved to understand their needs and tailor recommendations. Think of it as a team effort to optimise your cloud environment.
- ***Clear action plan:*** Review findings can be translated into a roadmap with steps to address each risk or improvement opportunity. This ensures everyone's on the same

Table 2: The pillars of AWS Well-Architected Review and their benefits

Pillar	Description	Benefits for cloud-native development in AWS
Operational excellence	Focuses on operational practices that enable continuous improvement of processes and procedures. This pillar emphasises automation, monitoring, and incident response.	Promotes the adoption of DevOps practices and automation in cloud-native development, streamlining processes and accelerating deployment cycles. Enables organisations to maintain high levels of operational efficiency, resilience, and agility in AWS environments.
Security	Addresses security best practices, including data protection, identity and access management, and compliance.	Ensures that cloud-native applications are built with robust security controls and measures in place, protecting sensitive data and mitigating risks associated with cyber threats and compliance requirements in AWS.
Reliability	Aims to ensure that systems can recover from failures and meet business requirements for availability.	Facilitates the design of highly available and fault-tolerant cloud-native applications in AWS, minimising downtime and ensuring continuous operation even in the face of failures or disruptions.
Performance efficiency	Focuses on optimising resource utilisation and maximising performance to meet business requirements.	Assists in improving the performance and scalability of cloud-native applications in AWS by optimising resource allocation, leveraging AWS services for caching and load balancing, and monitoring performance metrics to identify bottlenecks and areas for improvement.
Cost optimisation	Addresses strategies for optimising costs without sacrificing performance or reliability.	Helps in maximising cost efficiency in cloud-native development on AWS by identifying opportunities for cost optimisation, right-sizing resources, and leveraging cost-effective AWS services and pricing models.
Sustainability	Reusable energy in infrastructure services, GreenOps and carbon footprint calculation and reduction in cloud-native services.	Resource optimisation and cost benefits through efficient resource planning, and improves sustainability index in cloud infrastructure services.